

WOLF POINT 34 NE, MT, USA

WMO: 742310

Lat: 48.4887N

Lon: 105.2096W

Elev: 2643

StdP: 13.35

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 94059

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.7	-13.0	-23.3	1.7	-16.9	-18.7	2.2	-12.6	28.2	22.9	24.6	15.3	9.1	N/A	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	90.4	63.0	86.7	62.5	83.3	62.0	67.8	80.6	65.4	79.4	63.7	78.1	9.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.7	94.2	75.2	60.4	86.6	72.0	58.5	80.7	69.7	33.7	81.0	31.7	79.3	30.3	77.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	22.1	19.7	-23.5	96.6	3.0	4.0	-25.6	99.5	-27.4	101.8	-29.0	104.1	-31.2	107.0		
			-23.4	72.3	2.3	3.4	-25.0	74.7	-26.4	76.7	-27.7	78.6	-29.3	81.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	15.6	14.9	28.9	42.1	52.5	61.8	69.6	67.5	58.1	43.2	29.2	17.3
	DBStd	22.49	15.30	13.89	14.18	10.29	9.16	7.08	6.38	7.41	9.68	10.89	12.66	14.07
	HDD50	5001	1065	983	661	272	80	4	0	32	263	628	1013	
	HDD65	8843	1530	1403	1121	688	396	141	26	59	250	677	1074	1478
	CDD50	2042	0	0	5	35	158	359	608	543	276	54	4	0
	CDD65	406	0	0	0	0	9	46	169	136	44	2	0	0
	CDH74	4206	0	0	0	13	125	422	1712	1438	469	27	0	0
	CDH80	1503	0	0	0	1	20	102	661	558	158	3	0	0
	WSAvg	9.7	9.8	9.4	10.6	11.2	10.8	9.3	8.8	9.0	9.2	9.8	9.3	9.0
	PrecAvg	13.7	0.4	0.3	0.5	1.0	2.1	3.0	2.2	1.6	0.9	0.9	0.4	0.4
(15) Precipitation	PrecMax	20.4	1.2	0.7	1.6	2.2	5.9	6.1	8.0	6.2	2.5	3.1	1.7	0.9
	PrecMin	8.7	0.1	0.1	0.1	0.4	0.6	0.9	0.6	0.0	0.2	0.1	0.0	0.0
	PrecStd	2.4	0.3	0.2	0.3	0.5	1.4	1.2	1.4	1.3	0.7	0.7	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	46.4	48.1	66.8	75.5	82.7	88.3	96.0	95.8	90.7	77.8	62.6	47.4	
	0.4% MCWB	38.0	39.1	47.9	53.0	58.1	64.9	62.2	62.5	62.3	56.7	47.9	37.8	
	2% DB	41.5	41.7	58.3	68.3	77.5	82.9	90.7	90.4	84.5	70.5	55.2	42.3	
	2% MCWB	35.6	35.2	44.4	49.1	56.4	62.7	64.2	61.8	59.4	52.8	44.3	35.4	
	5% DB	38.4	37.1	52.5	63.4	73.3	78.6	87.1	86.2	79.7	64.2	50.2	38.7	
	5% MCWB	33.6	32.7	41.2	46.5	53.9	60.6	64.0	61.3	57.5	49.8	40.9	33.2	
	10% DB	35.3	33.4	47.1	58.8	68.6	75.3	83.4	82.2	74.3	59.2	45.5	35.1	
	10% MCWB	31.5	30.3	38.7	44.1	52.1	58.7	64.0	61.2	55.5	47.0	38.3	31.1	
	0.4% WB	39.2	40.3	48.2	53.8	62.2	70.1	72.7	68.2	65.8	57.8	48.7	39.0	
	0.4% MCDB	46.9	48.6	64.3	74.1	74.8	81.5	83.7	82.3	81.5	74.6	63.2	45.3	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	35.7	36.3	45.6	49.4	58.9	65.0	69.0	65.6	61.2	54.5	44.7	35.7	
	2% MCDB	41.1	42.4	59.1	64.8	72.1	78.1	82.2	81.0	78.2	66.6	55.1	42.1	
	5% WB	33.5	33.4	42.4	46.5	56.4	62.2	66.6	64.0	58.8	50.6	41.4	33.4	
	5% MCDB	37.7	37.4	52.3	60.1	69.8	75.0	79.4	79.6	73.9	61.7	49.5	38.5	
	10% WB	31.7	31.1	39.3	44.1	54.1	60.3	64.5	62.6	56.7	47.8	38.1	31.1	
	10% MCDB	35.3	34.4	47.4	56.3	66.3	72.4	78.1	77.8	71.8	58.5	44.3	34.9	
(35) Mean Daily Temperature Range	MDBR	14.9	15.0	16.9	19.9	20.1	19.1	22.0	22.1	20.7	17.8	15.6	13.5	
	5% DB	15.6	17.9	24.1	26.6	25.4	23.6	26.4	26.7	26.9	25.3	20.8	16.1	
	MCWBR	11.8	13.7	13.8	13.5	11.6	11.3	9.9	9.3	10.8	12.2	12.2	10.9	
	5% WB	15.8	17.7	24.2	25.3	22.1	20.9	22.8	23.8	24.0	23.1	20.9	15.8	
	MCWBR	12.1	13.4	14.1	13.4	11.1	11.5	11.3	9.4	10.7	11.5	12.5	11.6	
(40) Clear-Sky Solar Irradiance	taub	0.246	0.262	0.270	0.314	0.332	0.344	0.352	0.359	0.316	0.281	0.253	0.237	
	taud	2.465	2.462	2.548	2.430	2.430	2.440	2.435	2.384	2.473	2.564	2.535	2.499	
	Ebn at Noon	272	291	305	298	295	291	286	279	283	280	267	262	
	Edh at Noon	19	24	27	33	35	35	34	35	29	22	18	16	
(44) All-Sky Solar Radiation	RadAvg	459	788	1186	1568	1865	2024	2173	1777	1305	823	494	348	
	RadStd	40	52	74	70	145	117	104	123	120	83	56	20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page